SECTION 1 (Maximum marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
  Full Marks : $+3$ If **ONLY** the correct integer is entered;
  Zero Marks : 0 If the question is unanswered;
  Negative Marks : -1 In all other cases.

Q.1 Let α and β be real numbers such that $-\frac{\pi}{4} < \beta < 0 < \alpha < \frac{\pi}{4}$. If $\sin(\alpha+\beta) = \frac{1}{3}$ and $\cos(\alpha-\beta) = \frac{2}{3}$, then the greatest integer less than or equal to

$$\left(\frac{\sin\alpha}{\cos\beta} + \frac{\cos\beta}{\sin\alpha} + \frac{\cos\alpha}{\sin\beta} + \frac{\sin\beta}{\cos\alpha}\right)^2$$

is _____.

Q.2 If $y(x)$ is the solution of the differential equation

$$x dy - (y^2 - 4y)dx = 0 \quad \text{for} \quad x > 0, \qquad y(1) = 2,$$

and the slope of the curve $y = y(x)$ is never zero, then the value of $10\, y(\sqrt{2})$ is _____.

Q.3 The greatest integer less than or equal to

$$\int_1^2 \log_2(x^3+1)\, dx + \int_1^{\log_2 9} (2^x - 1)^{\frac{1}{3}} dx$$

is _____.

Q.4 The product of all positive real values of x satisfying the equation

$$x^{(16(\log_5 x)^3 - 68\log_5 x)} = 5^{-16}$$

is _____ .

Q.5 If

$$\beta = \lim_{x \to 0} \frac{e^{x^3} - (1-x^3)^{\frac{1}{3}} + ((1-x^2)^{\frac{1}{2}} - 1)\sin x}{x \sin^2 x},$$

then the value of 6β is _____.

Q.6 Let β be a real number. Consider the matrix

$$A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}.$$

If $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, then the value of 9β is _____.

Q.7 Consider the hyperbola

$$\frac{x^2}{100} - \frac{y^2}{64} = 1$$

with foci at S and S_1, where S lies on the positive x-axis. Let P be a point on the hyperbola, in the first quadrant. Let $\angle SPS_1 = \alpha$, with $\alpha < \frac{\pi}{2}$. The straight line passing through the point S and having the same slope as that of the tangent at P to the hyperbola, intersects the straight line S_1P at P_1. Let δ be the distance of P from the straight line SP_1, and $\beta = S_1P$. Then the greatest integer less than or equal to $\frac{\beta\delta}{9} \sin\frac{\alpha}{2}$ is _____.

Q.8 Consider the functions $f,\ g : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = x^2 + \frac{5}{12} \quad \text{and} \quad g(x) = \begin{cases} 2\left(1 - \frac{4|x|}{3}\right), & |x| \le \frac{3}{4}, \\ 0, & |x| > \frac{3}{4}. \end{cases}$$

If α is the area of the region

$$\left\{(x, y) \in \mathbb{R} \times \mathbb{R} : |x| \le \frac{3}{4},\ 0 \le y \le \min\{f(x), g(x)\}\right\},$$

then the value of 9α is _____.

SECTION 2 (Maximum marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated <u>according to the following marking scheme</u>:

Full Marks : $+4$ **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks : $+3$ If all the four options are correct but **ONLY** three options are chosen;
Partial Marks : $+2$ If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
Partial Marks : $+1$ If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
Zero Marks : 0 If unanswered;
Negative Marks : -2 In all other cases.

Q.9 Let $PQRS$ be a quadrilateral in a plane, where $QR = 1$, $\angle PQR = \angle QRS = 70^{o}$, $\angle PQS = 15^{o}$ and $\angle PRS = 40^{o}$. If $\angle RPS = \theta^{o}$, $PQ = \alpha$ and $PS = \beta$, then the interval(s) that contain(s) the value of $4\,\alpha\,\beta \sin\theta^{o}$ is/are

(A) $\left(0,\ \sqrt{2}\right)$

(B) $(1,\ 2)$

(C) $\left(\sqrt{2},\ 3\right)$

(D) $\left(2\sqrt{2},\ 3\sqrt{2}\right)$

Q.10 Let

$$\alpha = \sum_{k=1}^{\infty} \sin^{2k}\left(\frac{\pi}{6}\right) .$$

Let $g : [0,1] \to \mathbb{R}$ be the function defined by

$$g(x) = 2^{\alpha x} + 2^{\alpha(1-x)} .$$

Then, which of the following statements is/are TRUE ?

(A) The minimum value of $g(x)$ is $2^{\frac{7}{6}}$

(B) The maximum value of $g(x)$ is $1 + 2^{\frac{1}{3}}$

(C) The function $g(x)$ attains its maximum at more than one point

(D) The function $g(x)$ attains its minimum at more than one point

Q.11 Let $\bar{z}$ denote the complex conjugate of a complex number z. If z is a non-zero complex number for which both real and imaginary parts of

$$(\bar{z})^2 + \frac{1}{z^2}$$

are integers, then which of the following is/are possible value(s) of $|z|$?

(A) $\left(\frac{43+3\sqrt{205}}{2}\right)^{\frac{1}{4}}$

(B) $\left(\frac{7+\sqrt{33}}{4}\right)^{\frac{1}{4}}$

(C) $\left(\frac{9+\sqrt{65}}{4}\right)^{\frac{1}{4}}$

(D) $\left(\frac{7+\sqrt{13}}{6}\right)^{\frac{1}{4}}$

Q.12 Let G be a circle of radius $R > 0$. Let $G_1, G_2, \ldots, G_n$ be n circles of equal radius $r > 0$. Suppose each of the n circles $G_1, G_2, \ldots, G_n$ touches the circle G externally. Also, for $i = 1, 2, \ldots, n-1$, the circle G_i touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are TRUE ?

(A) If $n = 4$, then $(\sqrt{2} - 1)r < R$

(B) If $n = 5$, then $r < R$

(C) If $n = 8$, then $(\sqrt{2} - 1)\, r < R$

(D) If $n = 12$, then $\sqrt{2}\left(\sqrt{3} + 1\right) r > R$

Q.13 Let $\hat{\imath}, \hat{\jmath}$ and $\hat{k}$ be the unit vectors along the three positive coordinate axes. Let

$$\begin{aligned} \vec{a} &= 3\hat{\imath} + \hat{\jmath} - \hat{k}\,, \\ \vec{b} &= \hat{\imath} + b_2\hat{\jmath} + b_3\hat{k}\,, \qquad b_2, b_3 \in \mathbb{R}\,, \\ \vec{c} &= c_1\hat{\imath} + c_2\hat{\jmath} + c_3\hat{k}\,, \qquad c_1, c_2, c_3 \in \mathbb{R} \end{aligned}$$

be three vectors such that $b_2 b_3 > 0$, $\vec{a} \cdot \vec{b} = 0$ and

$$\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3 - c_1 \\ 1 - c_2 \\ -1 - c_3 \end{pmatrix}.$$

Then, which of the following is/are TRUE ?

(A) $\vec{a} \cdot \vec{c} = 0$

(B) $\vec{b} \cdot \vec{c} = 0$

(C) $|\vec{b}| > \sqrt{10}$

(D) $|\vec{c}| \leq \sqrt{11}$

Q.14 For $x \in \mathbb{R}$, let the function $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + 12y = \cos\left(\frac{\pi}{12}x\right), \qquad y(0) = 0\,.$$

Then, which of the following statements is/are TRUE ?

(A) $y(x)$ is an increasing function

(B) $y(x)$ is a decreasing function

(C) There exists a real number β such that the line $y = \beta$ intersects the curve $y = y(x)$ at infinitely many points

(D) $y(x)$ is a periodic function

SECTION 3 (Maximum marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : $+3$ If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.15 Consider 4 boxes, where each box contains 3 red balls and 2 blue balls. Assume that all 20 balls are distinct. In how many different ways can 10 balls be chosen from these 4 boxes so that from each box at least one red ball and one blue ball are chosen ?

(A) 21816 (B) 85536 (C) 12096 (D) 156816

Q.16 If $M = \begin{pmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}$, then which of the following matrices is equal to M^{2022} ?

(A) $\begin{pmatrix} 3034 & 3033 \\ -3033 & -3032 \end{pmatrix}$

(B) $\begin{pmatrix} 3034 & -3033 \\ 3033 & -3032 \end{pmatrix}$

(C) $\begin{pmatrix} 3033 & 3032 \\ -3032 & -3031 \end{pmatrix}$

(D) $\begin{pmatrix} 3032 & 3031 \\ -3031 & -3030 \end{pmatrix}$

Q.17 Suppose that

Box-I contains 8 red, 3 blue and 5 green balls,
Box-II contains 24 red, 9 blue and 15 green balls,
Box-III contains 1 blue, 12 green and 3 yellow balls,
Box-IV contains 10 green, 16 orange and 6 white balls.

A ball is chosen randomly from Box-I; call this ball b. If b is red then a ball is chosen randomly from Box-II, if b is blue then a ball is chosen randomly from Box-III, and if b is green then a ball is chosen randomly from Box-IV. The conditional probability of the event 'one of the chosen balls is white' given that the event 'at least one of the chosen balls is green' has happened, is equal to

(A) $\frac{15}{256}$ (B) $\frac{3}{16}$

(C) $\frac{5}{52}$ (D) $\frac{1}{8}$

Q.18 For positive integer n, define

$$f(n) = n + \frac{16 + 5n - 3n^2}{4n + 3n^2} + \frac{32 + n - 3n^2}{8n + 3n^2} + \frac{48 - 3n - 3n^2}{12n + 3n^2} + \cdots + \frac{25n - 7n^2}{7n^2} \ .$$

Then, the value of $\lim_{n \to \infty} f(n)$ is equal to

(A) $3 + \frac{4}{3}\log_e 7$ (B) $4 - \frac{3}{4}\log_e\left(\frac{7}{3}\right)$

(C) $4 - \frac{4}{3}\log_e\left(\frac{7}{3}\right)$ (D) $3 + \frac{3}{4}\log_e 7$

END OF THE QUESTION PAPER